



Cloud Native Observability

SLOSLASLI

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This article explains Service Level Objectives, Indicators, and Agreements, focusing on their roles in measuring and ensuring service reliability and performance.

When designing a robust system or application, it is essential for teams to define clear, measurable targets and goals. These targets allow organizations to strike a balance between product development and operational work while giving customers a concrete expectation of service reliability. For instance, an application might be required to deliver a minimum of 97% uptime over a rolling 30-day period.

A Service Level Indicator (SLI) is a quantitative metric that evaluates a specific aspect of service performance. In simpler terms, an SLI measures how well a service operates based on key performance indicators.

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When designing a system or applications, its important for teams to set specific measurable targets/goals to help organizations strike the right balance between product development and operation work.

These targets help customers & end users quantify the level of reliability they should come to expect from a service

"Application should have 97% uptime in a rolling 30 day window"

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Common SLIs include:

Request latency

Error rates

Saturation or throughput

Availability (e.g., ensuring 99% uptime)

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Service Level Indicator(SLI) – quantitative measure of some aspect of the level of service that is provided

Common SLIs:

- Request Latency
- Error Rate
- Saturation
- Throughput
- Availability

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It is important to understand that not every metric qualifies as an SLI. The most effective SLIs closely mirror the user's experience with the service. For example, high CPU usage or memory consumption may be a concern from a technical perspective but do not necessarily translate into a degraded user experience. Instead, metrics such as response time and error rates serve as more accurate indicators of service quality from the customer's viewpoint.

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Not all metrics make for good SLIs. You want to find metrics that accurately measure a **user's** experience.

Things like high-cpu/high-memory make for a poor SLI as a user might not see any impact on their end during these events



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A Service Level Objective (SLO) represents the target value or acceptable range for an SLI. For example, if an SLI reads latency, an SLO might be set to keep latency below 100 milliseconds. Similarly, an availability SLO could state that the service must achieve 99.9% uptime. Essentially, an SLO provides a clear, measurable goal aligned with the customer experience.

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Service Level Object(SLO) – target value or range for an SLI

SLI - Latency

SLO – Latency < 100ms

SLI – availability

SLO – 99.9% uptime

SLOs should be directly related to the **customer experience**. The main purpose of the SLO is to quantify reliability of a product to a customer

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While it might be tempting to push for extremely high targets such as 100% uptime or 99.999% uptime, this approach often leads to unnecessary costs without delivering significant benefits to customers.

A Service Level Agreement (SLA) is a formal contract between a vendor and a user that commits to meeting a specific SLO. SLAs are typically legally binding, and failure to meet the agreed-upon targets may result in financial penalties or other corrective measures.

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Service Level Agreement(SLA) – contract between a vendor and a user that guarantees a certain SLO



The consequences for not meeting any SLO can be financial based but can also be variety of other things as well

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In summary, SLIs are used to measure the quality of a service, SLOs set the target performance levels for these indicators, and SLAs formalize these targets in a binding agreement. Together, they play a critical role in ensuring that customers receive a consistent and high-quality service.

For further reading on service reliability management, explore our [Kubernetes Documentation](#) and [Docker Hub](#).



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